

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

May 2013

CL307 Structural Analysis-III

Date: 01.05.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Enlist losses in Pre-Stress and explain any two in detail. [07]

Q - 2 (a) Give advantages and disadvantages of Pre-Stressed Concrete. (Four Each) [04]

(b) Determine the support moments for the continuous beam shown in Figure 1 using moment distribution method. Draw bending moment diagram. [10]

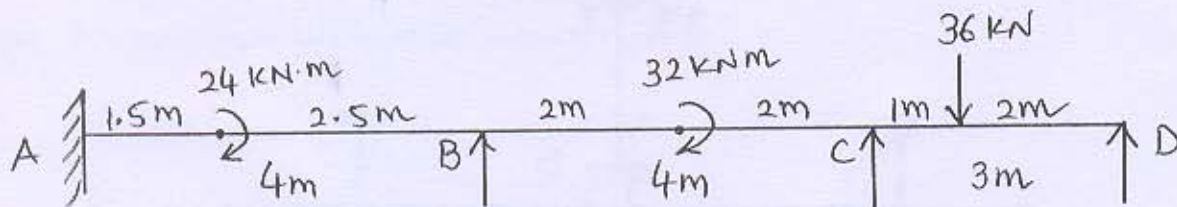


Figure 1

OR

(b) Analyse the portal frame shown in Figure 2 by Moment Distribution Method. Draw Bending moment diagram. [10]

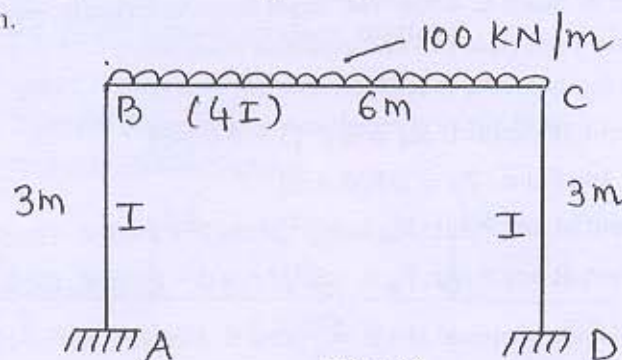


Figure 2

Q - 3 Using slope deflection method, determine the support moments and reactions for any two structures. Refer Figures 3, 4 & 5. Draw also B.M. Diagram. [14]

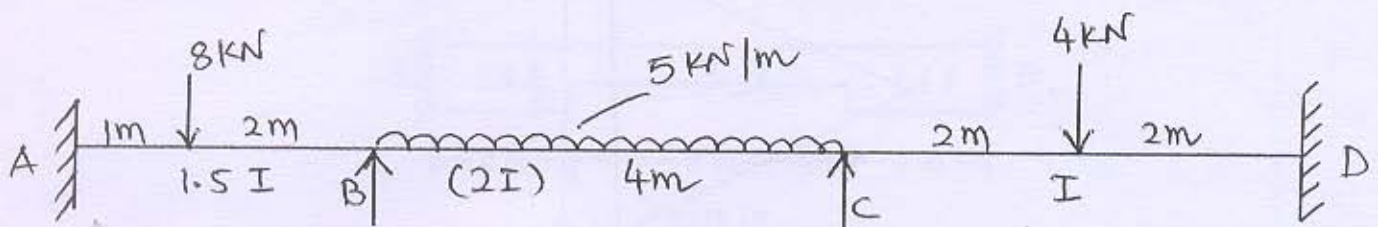


Figure 3

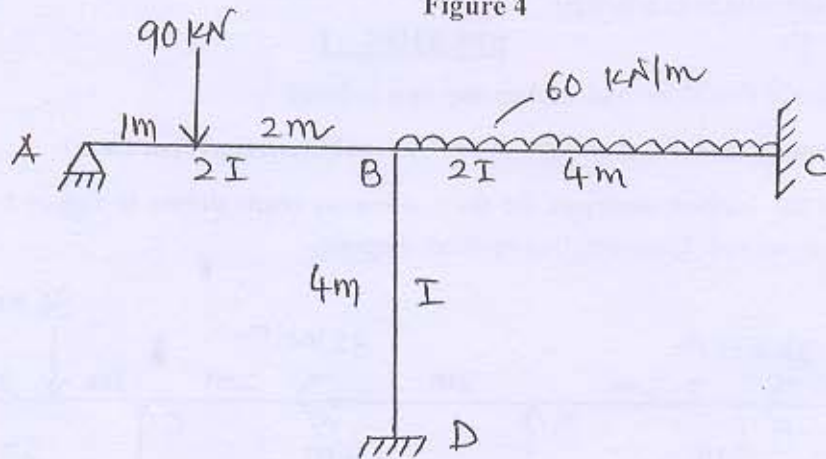
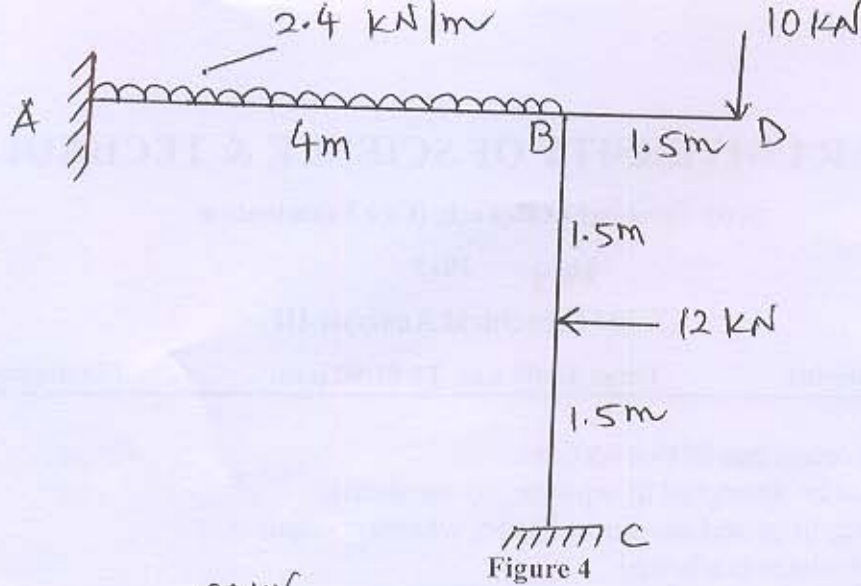


Figure 5
SECTION - II

Q - 4 A beam circular in plan is located uniformly with U.D.L. of 100 kN/m inclusive of self-weight. The radius of beam is 4.5m. The beam is symmetrically supported on five columns. Draw S.F., B.M. and T.M. (Use relevant equation from below) [07]

1. Shear Force at Support: $F_0 = wR\theta$
2. Bending Moment at Support: $M_0 = wR^2[1 - \theta \cot \theta]$
3. Shear Force at any Point : $F_\theta = wR[\theta - \phi]$
4. Bending Moment at any Point: $M_\theta = wR^2[\theta \sin \phi + \theta \cot \theta \cdot \cos \phi - 1]$
5. Twisting Moment at any Point: $T_\theta = wR^2[\theta \cos \phi - \theta \cot \theta \cdot \sin \phi - (\theta - \phi)]$
6. Position of Maximum Torque: $\sin \phi = \frac{1}{\theta}[\sin^2 \theta \pm \cos \theta \cdot (\theta^2 - \sin^2 \theta)^{\frac{1}{2}}]$

Q - 5 (a) Determine the shape factor for the triangular section shown in Figure 6. [05]

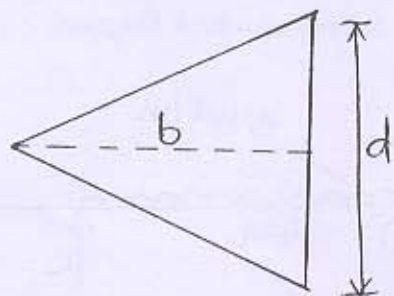


Figure 6

(b) Analyse the portal frame shown in Figure 7 by portal method/cantilever method.

[09]

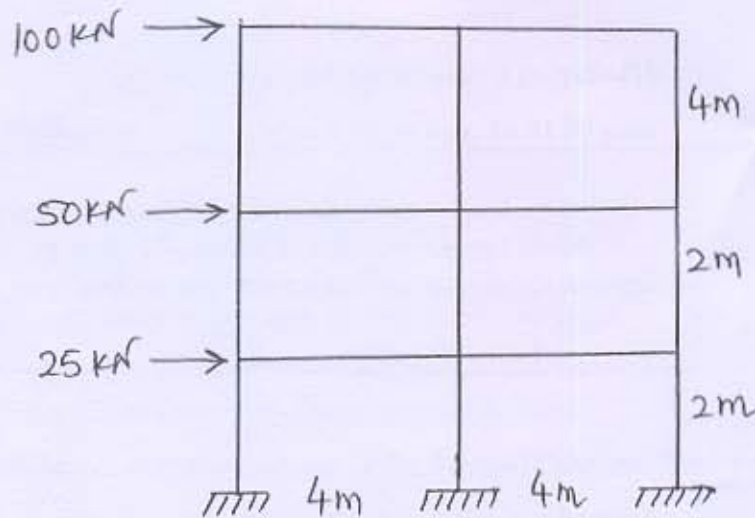


Figure 7

Q - 6 (a) Compute collapse load for a beam shown in Figure 8.

[04]

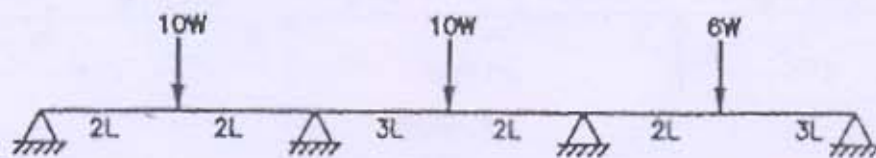


Figure 8

(b) Attempt any Two. (Use column analogy method)

[10]

- A fixed beam of span l carries a uniformly distributed load of w per unit run for a distance a from the left end. Find the fixing moment at the ends of the beam.
- Analyse the fixed beam shown in Figure 9.

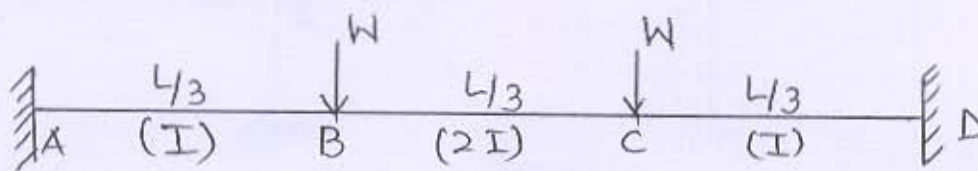


Figure 9

(iii) Find the stiffness and carry-over factor for the fixed beam shown in Figure 10.

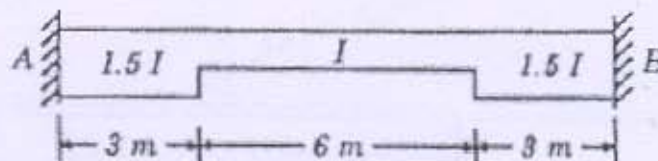


Figure 10
